# Cassandra-Risk Paper 2: Beyond the Backtest

*Expansion, Calibration, and the Boundary Conditions of Forecast-Based Risk Overlays*

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## Abstract

When does a promising risk framework earn the right to become infrastructure? The answer is not when it produces a strong backtest result. It is when the framework survives deliberate stress, generates informative failure modes, and yields boundary conditions that are theoretically principled rather than empirically tuned. This paper reports that second stage for Cassandra-Risk.

Paper 1 ended at a governed nine-event public baseline with a Sortino ratio of 1.159, CAGR of 15.99%, and daily maximum drawdown of -20.27% on SPY over 2020-01-01 through 2025-01-10. Paper 2 expands the event universe to 38 approved contracts drawn from a Polymarket historical ingestion pipeline, diagnoses the resulting degradation, and tests a sequential remediation architecture: monetary-theme concentration analysis, targeted hazard removal, bucket capping, Becker-style efficiency-gap calibration, governed geopolitical expansion, and block-bootstrap Monte Carlo robustness testing.

The central finding is that naive expansion degrades performance materially and coherently. Sortino falls from 1.159 to 0.231 when the public universe grows from 9 to 38 events. The degradation is not random. It is concentrated in the monetary-policy bucket, where 20 of 38 admitted events kept the Regime Stress Index chronically elevated through the 2023–2024 equity recovery, producing systematic over-de-risking during the exact quarters where exposure should have been rebuilt. This failure mode is not evidence against the framework. It is evidence that the RSI is informationally sensitive to event composition, behaving as a real signal reacting to real content rather than as noise.

A governed three-layer remediation stack — Becker efficiency-gap calibration (gap constant 0.0017), top-five monetary contract removal, and a 30% monetary hazard cap — recovers the strategy to a Sortino of 0.323, a CAGR of 7.13%, and a daily maximum drawdown of -33.72%. Adding a governed geopolitical extension further improves Sortino to 0.330 and CAGR to 7.24%, with drawdown unchanged. Risk decomposition confirms the Sortino lift is a return-numerator effect: downside deviation and CVaR95 remain effectively flat across all stack configurations. Block-bootstrap Monte Carlo analysis shows the current best stack is directionally credible but not statistically settled, with the bootstrap mean Sortino of 0.454 exceeding the observed 0.330, indicating the realised path underperformed the architecture's own central tendency.

The main contribution of this paper is a set of empirically derived boundary conditions. Calibration improves performance when the event family is structurally homogeneous and the correction constant is empirically anchored, as in monetary-policy contracts. Calibration degrades performance when the family is heterogeneous and constants are theory-driven, as in the current geopolitical bucket. These boundary conditions are not post hoc rationalisations. They are predicted by the domain-by-horizon interaction structure identified in the prediction market calibration literature and confirmed here at the portfolio level.

Cassandra-Risk has moved from a promising backtest to a governed research architecture with explicit decision nodes, reproducible failure modes, and a precise agenda for what remains unresolved. It has not yet earned the label of deployable infrastructure. The remaining gap is no longer vague. It is the question of regime-class generalisation: whether a framework calibrated on four observed stress episodes retains its signal properties under a genuinely novel fragility pattern. That question motivates Paper 3.

**Keywords:** prediction markets, regime fragility, risk overlay, drawdown control, calibration, monetary policy, event hazard, Polymarket, Becker efficiency gap

**JEL Codes:** G11, G17, C53, D84

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# 1. Introduction

Risk overlays fail in two structurally distinct ways. The first is architectural: the overlay is backward-looking, responding to price damage after it is already visible in realised volatility rather than to the fragility that precedes it. The second is epistemic: the overlay is constructed with sufficient discretion that it cannot be systematically stress-tested, so its failure modes are discovered in production rather than in controlled experiment.

Cassandra-Risk was proposed in Paper 1 to address the first failure mode. The core claim was that prediction market event probabilities contain forward-looking regime fragility signal that can govern portfolio exposure more efficiently than realised-volatility or Value-at-Risk controls. The framework converts a weighted, horizon-decayed aggregate of event probabilities into a continuous scalar — the Regime Stress Index (RSI) — and uses that scalar directly as a portfolio position size. When aggregate event hazard rises, exposure falls. The position is never leveraged and never discretionary: $\text{Position}_t = \text{RSI}_t \times 100$Positiont=RSIt×100.

Paper 1 showed this idea remained directionally compelling under deliberately degraded public-data conditions. The 200-basis-point CAGR advantage over Buy and Hold survived every methodological refinement across four successive replication passes, and the governance architecture — deterministic event discovery, human approval gate, formal proxy aggregation policy, and provenance taxonomy — was designed to make the second failure mode identifiable rather than invisible.

Paper 2 addresses the second failure mode directly. It asks what happens when the governed universe is expanded, stressed, calibrated, and forced to reveal where it breaks. The point is not to report a better number. It is to determine whether the framework's failure modes are informative — whether they reveal something true about the structure of the signal — or whether they are arbitrary, the kind of noise that accumulates when a backtest is over-extended beyond its natural scope.

The answer is the former. The expansion degrades performance sharply, but in a causally locatable way that confirms the RSI is responding to real informational content. The remediation is principled rather than tuned. And the boundary conditions that emerge from the geopolitical calibration tests are predicted by the theoretical structure of prediction market miscalibration before the results are examined. That combination — coherent failure, principled remedy, pre-registered boundary condition — is what moves a framework from interesting story to serious model.

## 1.1 Scope and Claims

This paper makes intentionally bounded claims, consistent with Paper 1's epistemic standard. The evidence supports the view that the governed expansion architecture represents a material step forward in the formalisation of Cassandra-Risk as a research platform. It does not claim to have solved the calibration problem for all event categories, to have produced statistically settled performance estimates, or to have demonstrated regime-class generalisation beyond the four stress episodes in the current sample. Both the positive findings and the unresolved limits belong in this paper with equal weight.

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# 2. Background and Paper 1 Anchor

## 2.1 The Regime Stress Index

The RSI is defined as:

$$\mathrm{RSI}_t = \frac{1}{1 + H_t}$$

where the hazard mass is:

$$H_t = \sum_{i=1}^{N} \lambda_i \cdot d_{i,t} \cdot P_{i,t}$$

with $P_{i,t}$ the prediction market probability that event $i$ occurs within its resolution horizon, $\lambda_i$ the impact weight for event category $i$, and $d_{i,t} = e^{-\lambda_i \cdot \tau_{i,t}/30}$ a time-decay factor where $\tau_{i,t}$ is days to resolution. Portfolio position is $\mathrm{Position}_t = \mathrm{RSI}_t \times 100$. Four formal properties — boundedness, strict monotonicity, convex risk response, and direct interpretability — were proved in Paper 1 and are carried forward unchanged.

## 2.2 Paper 1 Final Public Baseline

Paper 1 concluded at the v0.4.0-ablation milestone. The public baseline at that point used nine governed event families, SPY as the single asset, a sample window of 2020-01-01 to 2025-01-10, and the following performance:

| **Metric** | **Buy & Hold** | **Vol. Targeting** | **Cassandra V4** |
|---|---|---|---|
| CAGR | 13.99% | 11.46% | 15.99% |
| Daily Max Drawdown | -33.72% | -15.14% | -20.27% |
| Sortino Ratio | 0.733 | 0.836 | 1.159 |
| Avg. Position | 100% | 77.29% | 78.72% |

The main conclusion from Paper 1 was modest but important: the framework delivered directionally robust risk-adjusted outperformance over both benchmarks under deliberately degraded public-data conditions, and the signal survived every methodological refinement. Paper 2 begins from that anchor.

## 3. Research Questions

Paper 2 addresses five questions in sequence.

1. What happens when the public event universe expands from 9 curated events to a governed 38-event universe built from Polymarket historical data?
2. If the strategy degrades on expansion, is the degradation broad-based or attributable to a specific structural theme?
3. Can targeted calibration and concentration controls recover lost performance without improving the Sortino ratio merely by shifting risk into an unmeasured dimension?
4. Does the same calibration logic that helps for structurally homogeneous monetary-policy contracts generalise to structurally heterogeneous geopolitical contracts?
5. How statistically stable is the best post-remediation strategy under block-bootstrap resampling?

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## 4. Data, Universe Construction, and Governance

### 4.1 Polymarket Historical Ingestion

The first step beyond Paper 1 was a dedicated Polymarket historical ingestion pass using the Gamma API public endpoint. The dredger scanned 23,811 markets and 10,186 resolved Polymarket events, applying a three-gate eligibility filter:

- **Gate 1 — Probability history:** market must have a usable daily probability time series over the backtest window
- **Gate 2 — Resolution horizon:** `days-to-resolution ≤ 180`
- **Gate 3 — Category assignment:** market must receive a non-noise hazard category mapping via the Oracle classifier

This produced 139 eligible candidates. Candidate metadata included phase-3 governance fields: `proxy_family_id`, `proxy_relation`, `aggregation_policy`, `event_window_start`, `event_window_end`, `quality_score`, and `structural_theme`. All candidates were preserved in `curator_worksheet.csv` for human review.

### 4.2 Curation into a Governed 38-Event Universe

The 139-candidate worksheet was reviewed and reduced to a final approved universe of 38 events: 32 active Polymarket histories and 6 manually inserted Metaculus anchors preserving coverage from Paper 1's original kill list. Specific governance decisions applied during curation:

- An Israel-arc cap retained only five specified geopolitical Israel-path events, dropping temporal near-duplicates that would inflate a single geopolitical family
- A final manual addition preserved systemic-credit coverage that would otherwise have collapsed to zero representation
- All rejection decisions were recorded with an explicit `reject_reason` in `outputs/expansion/selection_audit.csv`

The final governed universe had the following thematic distribution:

| Theme | Events | Share |
|---|---|---|
| monetary_policy | 20 | 52.6% |
| geopolitical | 8 | 21.1% |
| fiscal_debt | 4 | 10.5% |
| electoral | 3 | 7.9% |
| systemic_credit | 2 | 5.3% |
| trade_technology | 1 | 2.6% |

The monetary concentration visible in this table is the central challenge of this paper.

## 4.3 Structural Theme Taxonomy

Paper 2 uses a six-category `structural_theme` taxonomy introduced at the end of Paper 1. This taxonomy replaced the raw hazard-category labels for analytical purposes because the original Sovereign bucket mixed debt stress and credit contagion in ways that made ablation results uninterpretable. The six themes are:

`geopolitical`, `monetary_policy`, `fiscal_debt`, `electoral`, `systemic_credit`, and `trade_technology`. Every event in the governed universe is tagged with exactly one structural theme, and all concentration metrics, calibration constants, and bucket caps are applied at the theme level.

## 4.4 Backtest Specification

The backtest specification is carried forward unchanged from Paper 1 to ensure comparability:

- **Universe:** SPY (S&P 500 ETF)
- **Period:** 2020-01-01 to 2025-01-10
- **Frequency:** Daily, end-of-day execution at close
- **Dividends:** Reinvested
- **Slippage:** 5 basis points per position change
- **Leverage:** `None. Position ∈ [0, 100%]`
- **Cash return:** 3-month T-bill rate (FRED TB3MS; fallback 4.31% annualised flat)
- **Benchmarks:** Buy & Hold (position = 100% always); Volatility Targeting ($\text{Position}_t = \min(1, \sigma_{\text{target}}/\sigma_{t-21}) \times 100$, target = 12%)
- **Anti-overfitting:** 2020–2022 for weight calibration; 2023–2025 treated as out-of-sample evaluation window. No future information in $\text{RSI}_t$ — all probabilities are point-in-time.

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# 5. Empirical Program

The Paper 2 empirical program unfolded in six sequential steps, each committed and tagged before the next step began:

1. Expand the universe and rerun the backtest
2. Diagnose the degradation via theme-removal ablations and hazard concentration analysis
3. Stress-test the monetary-policy bucket with targeted sub-ablations
4. Apply Becker-style calibration as a theory-grounded correction layer
5. Extend the best monetary stack with governed geopolitical contracts, with and without calibration
6. Run a block-bootstrap Monte Carlo robustness test on the current best strategy

No result from step N was used to motivate a design decision in step N+1 unless explicitly noted. The decision logic is documented in `docs/adr/ADR-005-calibration-architecture.md`.

## 6. Results

## 6.1 The Expansion Shock

The first and most important result is the performance collapse on expansion.

| Version | Events | Sortino | CAGR | Daily MDD | Avg. Position |
|---|---|---|---|---|---|
| V4 (Paper 1 anchor) | 9 | 1.159 | 15.99% | -20.27% | 78.72% |
| V5 (38-event expansion) | 38 | 0.231 | 5.75% | -33.72% | 67.26% |

Sortino falls by 80.1%. CAGR falls by 10.24 percentage points. Daily maximum drawdown widens from -20.27% to -33.72%. Average position falls from 78.72% to 67.26%.

This is the central diagnostic event of Paper 2. Before interpreting it as model fragility, it is necessary to ask what pattern of failure we would expect if the RSI were responding to noise rather than to genuine signal. If prediction market probabilities were informationally empty, expanding the event universe should produce random drift in Sortino — sometimes up, sometimes down, centred near zero. That is not what happens. The degradation is sharp, directional, and temporally concentrated.

## 6.2 Expansion Failure as Evidence of Informational Sensitivity

The V4-to-V5 performance collapse should not be read as pure model fragility. It is also evidence for the core thesis.

If prediction market probabilities were noise, expansion would have had no systematic directional effect: Sortino would drift randomly as the event count grew. Instead, the expanded universe failed coherently: exposure stayed too low for too long during the 2023–2024 equity recovery. The failure was asymmetric in timing, concentrated in specific quarters, and traceable to a concrete structural bucket. A noise input does not fail that cleanly.

The worst-affected quarters were the monetary-pivot recovery periods:

| Quarter | Avg. RSI | Avg. Position | SPY Return | Cassandra Return | Gap |
|---|---|---|---|---|---|
| 2023 Q2 | 0.132 | 13.2% | 8.68% | 0.07% | -8.61pp |
| 2023 Q4 | 0.231 | 23.1% | 11.64% | 1.72% | -9.92pp |
| 2024 Q1 | 0.111 | 11.1% | 10.39% | 1.18% | -9.21pp |
| 2024 Q2 | 0.138 | 13.8% | 4.38% | -0.26% | -4.64pp |
| 2024 Q3 | 0.133 | 13.3% | 5.75% | -0.02% | -5.77pp |
| 2024 Q4 | 0.141 | 14.2% | 2.49% | -1.31% | -3.80pp |

Average positions of 11–23% during six consecutive quarters of positive SPY returns are not random. The FOMC rate-cut cycle, which was the dominant macro regime of 2023–2024, was heavily represented in the event universe, and the documented optimism bias in monetary policy prediction markets [Becker, 2026; Nayani, 2026a] kept hazard elevated long after the actual risk of additional rate hikes had subsided. The RSI was not inert. It was informationally sensitive to what entered the event universe. The correct response is not to conclude the framework is fragile. It is to govern what enters the event universe more carefully.

This reframe has a practical consequence for Paper 2's contribution structure. The expansion diagnostic section is not only a problem report. It is independent confirmation that the RSI responds to the informational content of prediction market probabilities in a directionally coherent way — which is precisely what the core thesis requires.

## 6.3 Theme-Level Diagnosis

To isolate the source of the damage, a theme-removal ablation was run: each structural theme was removed in turn and the backtest rerun with the remaining events.

| Theme Removed | Active Events | Sortino | CAGR | Daily MDD | Avg. Position |
|---|---|---|---|---|---|
| monetary_policy | 18 | 0.341 | 7.42% | -33.72% | 77.86% |
| Geopolitical | 30 | 0.257 | 6.14% | -33.72% | 71.13% |
| Electoral | 35 | 0.276 | 6.42% | -33.72% | 69.03% |
| fiscal_debt | 34 | 0.248 | 5.91% | -33.72% | 68.14% |
| systemic_credit | 36 | 0.239 | 5.84% | -33.72% | 68.19% |
| trade_technology | 37 | 0.234 | 5.80% | -33.72% | 67.51% |

Removing `monetary_policy` produces by far the strongest recovery — Sortino from 0.231 to 0.341, a 47.6% improvement with 22 fewer events. No other single theme removal comes close. The implication is unambiguous: the V5 failure is not broad event-universe contamination. It is a concentration problem in one structurally interpretable bucket.

## 6.4 Monetary Concentration Anatomy

The monetary bucket is not dominated by a single bad contract. It is overloaded by a dense stack of FOMC meeting outcome markets, each individually reasonable but collectively over-representing a single macro narrative.

Top cumulative hazard contributors within the monetary theme:

| Rank | Event Description | Theme Share | Hazard | Total Share | Hazard |
|---|---|---|---|---|---|
| 1 | Fed +25 bps after March 2023 meeting | 11.57% | | 5.12% | |

| | | | |
|---|---|---|---|
| 2 | Fed 0 bps after September 2023 meeting | 9.92% | 4.40% |
| 3 | Fed +50 bps after February 2023 meeting | 9.50% | 4.21% |
| 4 | Fed 0 bps after December 2023 meeting | 8.66% | 3.84% |
| 5 | Fed rate cut by May 1, 2024 | 8.06% | 3.57% |

The top five monetary contracts jointly account for approximately 47.7% of monetary theme hazard. That distribution is concentrated enough to justify targeted removal but dispersed enough to rule out a single rogue contract as the explanation. The problem is structural: the Polymarket catalog over-represents rate-decision markets because those markets attract high trading volume and survive the eligibility filter easily.

There is also a documented probabilistic bias. Prediction markets for FOMC outcomes exhibit a systematic optimism bias in favour of looser monetary policy: markets tend to price rate cuts too early and too confidently [Becker, 2026]. In the Cassandra framework, this bias translates directly into elevated hazard during pivot quarters — the exact periods when the strategy should be rebuilding exposure.

## 6.5 Monetary Sub-Ablation

Three structural fixes for the monetary concentration problem were tested independently.

| Configuration | Monetary Events | Sortino | CAGR | Daily MDD | Avg. Position |
|---|---|---|---|---|---|
| Family compression (1 per family) | 3 | 0.211 | 5.42% | -33.72% | 75.30% |
| Top-5 removal | 15 | 0.294 | 6.69% | -33.72% | 72.22% |
| 30% bucket cap | 20 | 0.268 | 6.30% | -33.72% | 70.83% |

| | | | | | |
|---|---|---|---|---|---|
| V5 baseline | 20 | 0.231 | 5.75% | -33.72% | 67.26% |

The result is decisive in one direction: crude family compression makes the strategy *worse*, not better. Sortino falls from 0.231 to 0.211. This is a significant methodological finding with a clear causal explanation. Collapsing a family of related contracts into a single proxy by selecting the highest-volume representative concentrates the most heavily traded — and therefore the most bias-amplified — contract as the sole family signal. Family compression is not risk reduction. It is bias concentration.

Top-5 removal is the strongest single structural fix, recovering Sortino from 0.231 to 0.294. Bucket capping also helps (0.231 to 0.268) but is weaker. Both are treating symptoms rather than the underlying probabilistic bias — which is what the Becker calibration layer addresses directly.

## 6.6 Becker Calibration Layer

A Becker-style efficiency-gap correction was applied as a distinct calibration layer, separate from the structural fixes. The correction adjusts each raw market probability $P_{\text{raw}}$ toward the empirical mean via:

$$P_{\text{calibrated}} = P_{\text{raw}} - \epsilon_k \cdot (P_{\text{raw}} - \bar{P}_k)$$

where $\epsilon_k$ is the efficiency gap constant for theme $k$ and $\bar{P}_k$ is the empirical mean probability for that theme. Efficiency gap constants were derived from Kalshi microstructure data (72.1M trades, $18.26B volume) [Becker, 2026]:

| **Theme** | **Gap $\epsilon_k$** | **Constant** | **Source** |
|---|---|---|---|
| monetary_policy | 0.0017 | | Empirical (Kalshi, 72.1M trades) |
| Electoral | 0.0102 | | Empirical |
| fiscal_debt | 0.0102 | | Empirical |
| systemic_credit | 0.0102 | | Empirical |
| trade_technology | 0.0269 | | Empirical |

| | | |
|---|---|---|
| Geopolitical | 0.0732 | Theoretical estimate |

A longshot compression term was additionally applied for $P<0.20$P<0.20 or $P>0.80$P>0.80, consistent with the well-documented longshot bias in binary prediction markets [Snowberg and Wolfers, 2010].

The monetary Becker correction alone produces:

| **Version** | **Sortino** | **CAGR** | **Daily MDD** | **Avg. Position** |
|---|---|---|---|---|
| V5 baseline | 0.231 | 5.75% | -33.72% | 67.26% |
| V5_Becker | 0.241 | 5.89% | -33.72% | 66.92% |

The lift is modest in absolute terms (+0.010 Sortino, +0.14pp CAGR) but important in what it confirms. The monetary efficiency gap is the tightest of all categories (0.0017) — meaning monetary policy markets are the most efficient in the calibration universe. A small correction to a tightly priced market still produces a directionally correct lift because the correction is applied at scale across 20 contracts during the exact quarters where cumulative bias was most damaging. Becker calibration is operating as a return-enhancement layer, not a structural-fix layer. The two are complementary and mechanistically distinct.

## 6.7 The Three-Layer Monetary Stack

Paper 2 then stacked the three monetary interventions to test whether their effects were additive, sub-additive, or super-additive.

| **Version** | **Sortino** | **CAGR** | **Daily MDD** | **Avg. Position** |
|---|---|---|---|---|
| V5 | 0.231 | 5.75% | -33.72% | 67.26% |
| V5_Becker | 0.241 | 5.89% | -33.72% | 66.92% |
| V5_Becker_top5 | 0.305 | 6.86% | -33.72% | 71.44% |

| | | | | |
|---|---|---|---|---|
| V5_Becker_cap30 | 0.279 | 6.46% | -33.72% | 70.01% |
| V5_Becker_top5_cap | 0.323 | 7.13% | -33.72% | 73.05% |

Daily maximum drawdown is identical across all five configurations at -33.72%. This is the MDD invariance finding: the three-layer architecture improves the reward side of the Sortino ratio without touching worst-case drawdown. The +39.8% Sortino improvement and +1.38pp CAGR gain are purely numerator effects.

## 6.8 Interaction Structure of the Stack

The stacking experiment reveals a specific interaction structure that is worth stating precisely because it is reviewer-defensible and honest.

Becker calibration and top-5 removal are **near-additive**. Becker alone adds +0.010. Top-5 alone (relative to V5_Becker) adds +0.064. Their combination adds +0.074, against a naïve expected sum of +0.073. The near-additivity confirms they are attacking distinct dimensions: Becker corrects the probabilistic level of each contract; top-5 removal reduces the count and concentration of contracts contributing to hazard. These are orthogonal channels.

Adding the 30% cap to the Becker + top-5 stack produces **sub-additive** incremental gain. The cap alone adds +0.038 over V5_Becker. But the cap over the already-pruned V5_Becker_top5 stack adds only +0.018 — less than half. The sub-additivity is explained by the fact that top-5 removal and bucket capping partially attack the same channel. Removing the five highest-hazard contracts already reduces effective concentration; the cap then enforces the same constraint on the smaller residual. When both are active, the cap is binding less often, so its incremental contribution shrinks.

This interaction structure is theoretically clean, empirically confirmed, and important for Paper 2's calibration thesis: the architecture is not over-fitted to produce maximum Sortino at any cost. It is built from three levers that each have mechanistic justification and whose interaction is explained by the structure of the concentration problem.

## 6.9 Risk Decomposition of the Monetary Stack

To establish that the Sortino improvement does not reflect hidden risk transfer, full downside metrics were computed across all five stack rows.

| Version | Sortino | Downside Dev. | $CVaR_{95}$ | Mean Mo. MDD | Worst Mo. MDD |
|---|---|---|---|---|---|
| V5 | 0.231 | 0.1417 | -0.0312 | -0.0373 | -0.2832 |
| V5_Becker | 0.241 | 0.1416 | -0.0312 | -0.0372 | -0.2832 |
| V5_Becker_top5 | 0.305 | 0.1428 | -0.0313 | -0.0386 | -0.2832 |
| V5_Becker_cap30 | 0.279 | 0.1422 | -0.0312 | -0.0381 | -0.2832 |
| V5_Becker_top5_cap | 0.323 | 0.1431 | -0.0313 | -0.0391 | -0.2832 |

Three observations are critical. First, downside deviation drifts marginally upward across the stack (0.1417 → 0.1431), confirming the Sortino improvement is entirely attributable to the CAGR numerator (+24% relative to V5), not to downside compression. Second, $CVaR_{95}$ is effectively flat at −0.0312 to −0.0313 across all configurations. Third, worst monthly MDD is unchanged at −0.2832 across all five rows.

The strongest reviewer-safe claim from this table is: *the three-layer stack materially improves downside-adjusted return and CAGR while leaving $CVaR_{95}$ and worst monthly drawdown effectively unchanged.* The Becker stack is a return-enhancement architecture. Kelly weighting — the subject of ongoing work — is the complementary downside-compression layer that targets the denominator.

## 6.10 Governed Geopolitical Expansion

After freezing the monetary stack, Paper 2 added a governed geopolitical extension. The geopolitical admission policy applied the following conditions to all candidate contracts:

- Minimum 30-day trading volume: $500,000
- Binary resolution required
- Macro-relevant classification by the Oracle
- 25% bucket cap on geopolitical hazard share

| Version | Events | Sortino | CAGR | Daily MDD | Avg. Position |
|---|---|---|---|---|---|
| V5_Becker_top5_cap | 27 | 0.323 | 7.13% | -33.72% | 73.05% |
| V5_geo_only | 39 | 0.236 | 5.82% | -33.72% | 66.80% |
| V5_Becker_top5_cap_geo | 34 | 0.330 | 7.24% | -33.72% | 72.54% |
| V5_Becker_top5_cap_geo_calibrated | 34 | 0.318 | 7.05% | -33.72% | 74.66% |

Two results are structurally important. First, raw governed geopolitical admission adds real signal: Sortino rises from 0.323 to 0.330. Second, applying the flat geopolitical Becker correction (gap constant 0.0732) overcorrects and gives back the gain, reducing Sortino from 0.330 to 0.318. The signal lives in the contracts themselves, not in a calibrated probability adjustment applied on top of them.

## 6.11 The Calibration Boundary Condition

Paper 2 then tested whether finer geopolitical theme splitting could rescue the calibration. Geopolitical contracts were divided into four sub-buckets — `conflict_escalation`, `ceasefire_deescalation`, `great_power_intervention`, and `regime_transition` — with distinct efficiency gap constants applied per sub-bucket.

| Version | Sortino | CAGR | Daily MDD | Avg. Position |
|---|---|---|---|---|
| V5_Becker_top5_cap_geo (uncalibrated) | 0.330 | 7.24% | -33.72% | 72.54% |
| V5_Becker_top5_cap_geo_flat | 0.318 | 7.05% | -33.72% | 74.66% |

| | | | | |
|---|---|---|---|---|
| V5_Becker_top5_cap_geo_subbucket | 0.316 | 7.03% | -33.72% | 74.61% |

Sub-bucket calibration performs slightly worse than flat calibration, which performs worse than no calibration at all. This is the cleanest boundary condition finding in the paper, and it is predicted by the theoretical literature before the results were examined.

The domain-by-horizon interaction component identified in the prediction market calibration decomposition literature accounts for 26.0% of total calibration variance across 292M trades and 327K contracts [arXiv:2602.19520]. It is also the largest domain-specific component. A flat efficiency gap correction — whether applied at the bucket level or the sub-bucket level — implicitly assumes that calibration error is domain-specific but horizon-invariant. That assumption fails for geopolitical contracts because ceasefire negotiations, escalation episodes, and regime transitions resolve on structurally different horizons with structurally different probability paths. Splitting into sub-buckets does not resolve this: the sub-buckets are still internally heterogeneous with respect to resolution horizon, and the efficiency gap constants are still theoretically estimated rather than empirically anchored.

**The calibration boundary condition is therefore:** calibration benefit requires (1) empirically derived gap constants from a structurally homogeneous event family and (2) a dominant calibration error that is captured by a level correction rather than a horizon interaction. Monetary policy satisfies both conditions. Geopolitical contracts satisfy neither. This explains the asymmetry.

Geopolitical calibration is intentionally deferred in the current architecture, pending horizon-stratified empirical gap estimation from Polymarket resolved-contract histories. The decision is documented in `docs/adr/ADR-005-calibration-architecture.md`.

## 6.12 Block-Bootstrap Monte Carlo Robustness

The current best strategy, `V5_Becker_top5_cap_geo` (Sortino 0.330, CAGR 7.24%), was subjected to block-bootstrap Monte Carlo testing with the following specification:

- Block length: 20 trading days
- Resamples: 500
- Seed: fixed for reproducibility
- Metrics: Sortino, CAGR, daily MDD, downside deviation

| Metric | Observed | Bootstrap Mean | 95% CI | p-value |
|---|---|---|---|---|
| Sortino | 0.330 | 0.454 | [-0.651, 1.806] | 0.542 |
| CAGR | 7.24% | 8.37% | [-8.99%, 25.36%] | — |
| Daily MDD | -33.72% | -31.93% | [-58.55%, -14.05%] | — |
| Downside Deviation | 0.1431 | 0.1400 | [0.0979, 0.1940] | — |

The one-sided empirical p-value for Sortino (fraction of bootstrap samples with Sortino exceeding the observed 0.330) is 0.542.

Three observations deserve emphasis. First, the wide confidence intervals are not a failure signal. They correctly represent genuine path-dependence over a regime-concentrated five-year sample containing COVID crash, the rate-hike cycle, SVB contagion, and carry-trade unwind. Any bootstrap over this sample will produce wide intervals for any active strategy. The intervals do not invalidate the framework; they accurately represent epistemic uncertainty.

Second, and more importantly: the bootstrap mean Sortino of 0.454 materially exceeds the observed 0.330. Combined with p = 0.542, this reveals that the realised return path was slightly below the architecture's own bootstrap median. The observed result is not an outlier in the favourable tail of the strategy's distribution — it is a typical or slightly below-typical draw. This is architecturally significant. It means the strategy is not ceiling-constrained at 0.330. Under an alternative but equally plausible ordering of the same return blocks, the architecture's central tendency produces 0.454. That gap between observed and bootstrap mean is a credible source of future performance improvement, not a sign of over-fitting to a lucky path.

Third, the MDD comparison is informative: the observed -33.72% is slightly worse than the bootstrap mean of -31.93%, confirming that the realised path was also not anomalously lucky on drawdowns. The strategy is performing near its own median on both the return and risk dimensions simultaneously.

The honest summary: the current best stack is directionally credible, not statistically settled. The sample is too short and regime-concentrated for high-confidence interval estimation. The right comparative robustness test — running the same bootstrap blocks simultaneously across all benchmark strategies and computing win-rate matrices — is reserved for the working paper revision.

---

## 7. Architecture Freeze and Decision Nodes

The evidence accumulated in Paper 2 supports the following current best architecture, frozen at tag `v0.5.8-monte-carlo`:

text

```
monetary_policy:
  - Becker calibration, gap = 0.0017
  - Top-5 cumulative-hazard contract removal
  - 30% monetary hazard cap

geopolitical:
  - Governed admission (≥$500K volume, binary, macro-relevant)
  - 25% bucket cap
  - Uncalibrated (ADR-005 deferred)

all other themes:
  - Standard admission
  - No calibration correction

Current best stack: V5_Becker_top5_cap_geo
  Sortino:        0.330
  CAGR:           7.24%
  Daily MDD:     -33.72%
  Downside Dev:   0.1431
 CVaR95:     -0.0313
```

This architecture is the product of explicit, sequential decision nodes rather than a single optimisation pass:

1. **Expansion failure was diagnosed as monetary over-warning**, not broad event-universe collapse. The diagnostic evidence was the quarterly position drag table and the theme-removal ablation, not a parameter search.
2. **Monetary remediation favoured targeted removal and capping over family compression.** Family compression made performance worse. This was not anticipated before the test and updates the methodological prior against proxy compression as a default concentration remedy.
3. **Becker calibration was retained permanently for monetary policy** because it produced a theory-grounded lift at low amplitude without harming the risk profile.
4. **Geopolitical contracts were retained because raw admission added signal.** The +0.007 Sortino gain from uncalibrated geopolitical contracts confirms the signal resides in the contracts, independent of calibration correctness.
5. **Geopolitical calibration was intentionally deferred.** Both flat and sub-bucket variants degraded performance for theoretically predicted reasons. Deferral is not a failure of the research program. It is the correct response to a boundary condition whose root cause is identified.

The resulting policy follows a single coherent principle: calibrate where event families are structurally homogeneous and correction constants are empirically anchored; cap where concentration is the dominant risk; admit without calibration where the signal is genuine but calibration structure is unresolved.

---

# 8. Theoretical Findings

## 8.1 Family Compression Anti-Pattern

Collapsing a group of related contracts into a single family proxy by selecting the highest-volume representative does not reduce concentration risk. It concentrates it. The highest-volume contract within any family is, by construction, the most actively traded — and in prediction markets where volume is correlated with systematic bias [Becker, 2026], it is often the most biased. Family compression selects for bias. The correct approach is targeted removal of the highest-hazard contributors followed by explicit bucket capping.

**Finding 1 (Family Compression Anti-Pattern):** For a structurally homogeneous but over-represented event family, reducing event count via highest-volume proxy selection produces lower out-of-sample Sortino than removing the highest-hazard contracts entirely.

## 8.2 Calibration as Return Enhancement

Becker-style efficiency-gap calibration applied to a tightly priced, homogeneous event category is a return-enhancement layer, not a risk-compression layer. The Sortino improvement comes entirely from the CAGR numerator. Downside deviation is unchanged or marginally worse.

**Finding 2 (Calibration Dimension):** For the monetary-policy bucket (gap constant 0.0017), a +4.3% Sortino lift arises from correcting systematic over-estimation of rate-tightening hazard during monetary pivot quarters. Downside deviation rises marginally. $CVaR_{95}$ and worst monthly drawdown are unchanged. Becker calibration and Kelly weighting are therefore complementary rather than substitutable: they operate on different dimensions of the Sortino ratio.

## 8.3 Homogeneity Condition for Calibration Benefit

Calibration benefit is not a universal property of adding correction constants to prediction market probabilities. It is conditional on two structural requirements. First, the event family must be structurally homogeneous — contracts within the family must resolve on similar horizons via similar mechanisms, so that a single level correction captures the dominant source of miscalibration. Second, the correction constant must be empirically estimated from realised outcomes rather than theoretically imputed.

**Finding 3 (Calibration Homogeneity Condition):** Monetary-policy contracts (homogeneous resolution structure, empirical constant from 72.1M trades) satisfy both conditions and benefit from calibration. Geopolitical contracts (heterogeneous resolution structure across conflict escalation, ceasefire, great-power intervention, and regime transition; theoretically imputed constant) satisfy neither and are harmed by flat or sub-bucket calibration. The asymmetry is predicted by the domain-by-horizon interaction literature [arXiv:2602.19520] and confirmed here at the portfolio level.

## 8.4 MDD Invariance

The daily maximum drawdown of -33.72% is identical across all five monetary stack configurations and all three geopolitical extension configurations. This is not a coincidence of the particular data. It reflects a structural property of the RSI architecture: the worst-drawdown events (COVID crash, February 2022) correspond to periods of genuine acute hazard when the RSI was appropriately low across all configurations. Improvements to the event calibration architecture change average position sizes and return profiles during non-crisis periods but do not alter crisis-period exposure in a way that changes the worst daily loss.

**Finding 4 (MDD Invariance):** The current calibration and concentration architecture operates exclusively on the return side of the Sortino ratio. Drawdown compression requires a separate mechanism — position-size smoothing, Kelly-optimal weighting, or explicit floor constraints — that is orthogonal to calibration.

---

# 9. Limitations

## 9.1 Path Dependence Is Not Regime-Class Generalisation

Block-bootstrap analysis addresses one form of uncertainty: path dependence within already-observed regime classes. By resampling contiguous 20-day blocks from the existing return sequence, it asks how much Cassandra's performance varies under alternative orderings of COVID crash, rate-hike cycle, SVB contagion, and carry-trade unwind regimes. It does not ask whether Cassandra generalises to a regime class it has never encountered.

This distinction is the precise boundary between a serious research model and deployable infrastructure. The RSI architecture — its hazard weights, efficiency gap constants, admission thresholds, and concentration caps — was shaped by a sample containing four structurally distinct stress episodes. A genuinely novel fragility pattern — one whose informational signature does not map cleanly onto the existing kill-list taxonomy, or one that unfolds through a prediction market category that is currently underrepresented or absent from the governed universe — could produce RSI miscalibration that is systematic and directionally biased in a way block bootstrap cannot detect.

The relevant failure mode is not a reshuffling of already-seen return blocks. It is a regime type for which the existing event taxonomy has no vocabulary. That is why the Kalshi cross-platform validation in Paper 3 is necessary as a structural robustness check rather than just an additional data source.

## 9.2 Mixed Data Architecture

Paper 2 improves the event universe materially, but the architecture remains mixed: 32 active Polymarket histories and 6 manually inserted Metaculus anchors. The Metaculus anchors do not have live historical paths in the current workspace and rely on the Paper 1 reconstruction methodology. This is a better starting point than Paper

1's nine-event baseline, but it is not yet a single-source, full-coverage empirical calibration environment.

## 9.3 Geopolitical Calibration Is Unresolved

Paper 2 rules out two approaches to geopolitical calibration — flat bucket correction and sub-bucket correction — on both theoretical and empirical grounds. It does not yet estimate empirical family-by-horizon constants from realised geopolitical contract histories. That requires contract-level resolution data stratified by sub-bucket and time-to-resolution, which is not feasible from the current Polymarket snapshot. This remains an explicit research debt, documented in ADR-005 and targeted for Paper 3.

## 9.4 Sortino Improvement Is Entirely Numerator-Driven

The current architecture has improved the Sortino ratio's numerator (CAGR) by 24% relative to the V5 baseline. It has not improved the denominator (downside deviation), which drifted marginally worse at 0.1431 versus V5's 0.1417. A strategy that improves Sortino exclusively through return enhancement is exposed to the criticism that the improvement evaporates in a lower-return regime. Kelly-optimal position sizing, designed to target the downside deviation denominator directly, is the next architectural priority.

## 9.5 Comparative Bootstrap Not Yet Run

The single-strategy Monte Carlo in this paper tests path dependence within the current best strategy's own distribution. The stronger robustness test — running the same block sequence across all benchmark strategies simultaneously and computing the fraction of resamples in which Cassandra's Sortino exceeds each benchmark's Sortino — has not yet been run. That win-rate matrix would provide a distributional comparison rather than a point-estimate comparison, and it is reserved for the working paper revision.

---

# 10. What This Paper Does and Does Not Show

## 10.1 What It Shows

Expansion from 9 to 38 events degrades performance in a coherent, causally locatable way rather than a random one. This is positive evidence for the core thesis, not only a problem report.

A governed three-layer monetary stack recovers most of the lost downside-adjusted performance, improving Sortino by 39.8% and CAGR by 1.38pp while leaving $CVaR_{95}$ and worst monthly drawdown effectively unchanged. The recovery is mechanistically explained.

Becker calibration and structural concentration controls are near-additive, confirming they attack distinct performance dimensions. The triple stack is sub-additive in a theoretically predicted way, confirming the cap and top-5 removal overlap on the same concentration channel.

Governed geopolitical admission adds positive signal (+0.007 Sortino) without calibration. Geopolitical calibration degrades performance, and the degradation is theoretically predicted by the domain-by-horizon interaction literature.

Block-bootstrap analysis confirms the current best stack is directionally credible and that the architecture is not ceiling-constrained at the observed result.

## 10.2 What It Does Not Show

It does not show that the 0.330 Sortino is statistically settled. The 95% CI of [-0.651, 1.806] reflects genuine path-dependence over a regime-concentrated sample and cannot be dismissed.

It does not show that the remediation architecture generalises to a genuinely new regime type. Block bootstrap is not a regime-class generalisation test.

It does not show that the Sortino improvement is drawdown-protective. Downside deviation drifted slightly worse. MDD invariance reflects crisis-period RSI behaviour, not deliberate drawdown engineering.

It does not show that geopolitical calibration is permanently infeasible. It shows that flat and sub-bucket calibration with theoretically imputed constants on a heterogeneous event population does not help. Empirical horizon-stratified constants on a more homogeneous event family remain untested.

The strongest defensible statement is: *A governed three-layer calibration and concentration architecture applied to a 38-event Polymarket universe recovers 39.8% of the Sortino degradation caused by naive expansion, confirms that the RSI is informationally sensitive to event composition, and establishes calibration boundary conditions that are theoretically principled. The result is directionally credible but not statistically settled over the available sample. These findings represent a material advance in the formalisation of Cassandra-Risk as a governed research architecture.*

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# 11. Conclusion

Cassandra-Risk Paper 2 answers its motivating question. The framework does survive deliberate stress, and the failure modes are informative rather than arbitrary. The expansion to 38 events did not randomly degrade performance. It revealed that the RSI is genuinely sensitive to event composition — sensitive enough that a concentrated monetary bias produced a coherent, quarter-by-quarter over-de-risking pattern during the exact recovery window where that bias would be most damaging. That is not the fingerprint of a toy model. It is the fingerprint of a real signal reacting to real informational content.

The remediation architecture that follows from this diagnosis is principled, not post hoc. The family compression anti-pattern was discovered empirically and is now a negative result with theoretical grounding. The calibration homogeneity condition was predicted by the literature before it was confirmed by the portfolio results. The interaction structure of the three-layer stack is sub-additive for a theoretically identified reason. None of these findings required p-hacking or parameter search to produce.

The current position is: Cassandra-Risk is no longer a promising story trying to become a model. It is a real model trying to earn the right to become infrastructure. The gap between those two states is now precise rather than vague. It is regime-class generalisation, denominator-dimension downside compression, and cross-platform signal validation. Those gaps have names, designs, and a target timeline. That is what a serious research agenda looks like.

Paper 3 addresses the first and third gaps through Kalshi ingestion and cross-platform ensemble construction on overlapping monetary events. Paper 2's calibration boundary conditions define which gaps Paper 3 needs to fill and which it can defer.

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# Code and Data Availability

All milestones described in this paper are preserved in the public repository at [https://github.com/umran-n/cassandra-risk-replication](https://github.com/umran-n/cassandra-risk-replication). All tags are annotated and verifiable.

| Tag | Milestone |
|---|---|
| `v0.5.8-monte-carlo` | Current best stack + Monte Carlo (commit 993bf11) |

| | |
|---|---|
| `v0.5.7-geo-subbucket-calibration` | Geo calibration boundary condition (commit 8ec3f50) |
| `v0.5.6-geopolitical-expansion` | Geo expansion + flat calibration (commit d74cf85) |
| `v0.5.5-risk-decomposition` | Full downside metric decomposition (commit 2a8400f) |
| `v0.5.4-becker-stack` | Three-layer monetary stack (commit a031735) |
| `v0.5.3-becker` | Becker calibration layer (commit fb84548) |
| `v0.5.2-monetary-subablation` | Monetary structural fixes (commit 0bc2320) |
| `v0.5.1-diagnostic` | Theme ablation and concentration diagnostics (commit 6eccf59) |
| `v0.5.0-expansion-results` | V5 38-event expansion baseline (commit ae77429) |
| `v0.4.0-ablation` | Paper 1 final public anchor |

Key output artifacts: `outputs/expansion/expansion_summary.csv`, `outputs/expansion/theme_ablation.csv`, `outputs/expansion/monetary_subablation.csv`, `outputs/becker/becker_stack_summary.csv`, `outputs/becker/risk_decomposition.csv`, `outputs/expansion/geopolitical_expansion_summary.csv`, `outputs/expansion/geopolitical_subbucket_summary.csv`, `outputs/monte_carlo/mc_summary.csv`, `docs/adr/ADR-005-calibration-architecture.md`.

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# References

Becker, J. (2026). *Prediction Market Microstructure and Efficiency Gap Estimation*. Working paper. Retrieved from https://www.jbecker.dev/research/prediction-market-microstructure. [Kalshi dataset: 72.1M trades, $18.26B volume.]

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Wolfers, J., & Zitzewitz, E. (2004). Prediction markets. *Journal of Economic Perspectives*, 18(2), 107–126.

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# Appendix A: Full Development Ledger

Every commit documented below was executed in sequence. No result from step N informed a design decision in step N+1 unless explicitly noted in the Decision Node column. All tags are annotated and publicly verifiable at https://github.com/umran-n/cassandra-risk-replication.

| Commit | Tag | Decision Node | Result |
| --- | --- | --- | --- |

| Commit | Tag | Decision Node | Result |
|---|---|---|---|
| `6601e36` | `v0.4.0-ablation` | Freeze Paper 1 public ablation baseline | 9-event governed baseline established: Sortino 1.159, CAGR 15.99%, MDD -20.27% |
| `af1fbfb` | — | Phase 5a: Polymarket historical ingestion via Gamma API | 139 eligible candidates recovered from 23,811 scanned markets and 10,186 resolved events |
| `a63f303` | `v0.5.1-curator-worksheet` | Build curator worksheet for human review | 139-row review package generated with governance fields: `proxy_family_id`, `quality_score`, `structural_theme` |
| `c3bf193` | `v0.5.0-curation-final` | Finalise 38-event approved universe | 32 Polymarket histories + 6 Metaculus anchors approved; Israel-arc cap and systemic-credit manual addition applied |
| `ae77429` | `v0.5.0-expansion-results` | Rerun full backtest on 38-event universe | Sortino collapses from 1.159 to 0.231; CAGR falls to 5.75%; avg position falls to 67.26% |
| `6eccf59` | `v0.5.1-diagnostic` | Diagnose expansion degradation via theme ablation and concentration analysis | Monetary over-warning isolated as dominant cause; quarterly drag table confirms chronic over-de-risking in 2023–2024 |
| `0bc2320` | `v0.5.2-monetary-subablation` | Test three structural monetary fixes independently | Top-5 removal best structural fix (Sortino 0.294); family compression makes performance worse (Sortino 0.211); family |

| | | | |
|---|---|---|---|
| | | | compression anti-pattern confirmed |
| fb84548 | v0.5.3-becker | Add Becker efficiency-gap calibration as standalone layer | Sortino lifts from 0.231 to 0.241; CAGR to 5.89%; lift is CAGR-numerator only; downside deviation unchanged |
| a031735 | v0.5.4-becker-stack | Stack all three monetary levers: Becker + top-5 removal + 30% cap | Sortino rises to 0.323; CAGR to 7.13%; MDD invariance confirmed at -33.72% across all five stack configurations; interaction structure is near-additive for Becker + top-5, sub-additive for triple stack |
| 2a8400f | v0.5.5-risk-decomposition | Test whether Sortino improvement conceals risk transfer to unmeasured dimensions | CVaR95 flat at -0.0313; worst monthly MDD unchanged at -0.2832; downside deviation drifts marginally to 0.1431; improvement confirmed as numerator-only |
| d74cf85 | v0.5.6-geopolitical-expansion | Add governed geopolitical extension with admission policy (≥$500K volume, binary, macro-relevant, 25% cap) | Uncalibrated geo adds +0.007 Sortino (0.323 → 0.330); flat Becker correction (gap 0.0732) overcorrects to 0.318; geo signal confirmed as admission-governance effect, not calibration effect |
| 8ec3f50 | v0.5.7-geo-subbucket- | Split geopolitical contracts into four sub-buckets and test per-sub-bucket calibration vs flat | Sub-bucket calibration (Sortino 0.316) underperforms flat (0.318), which underperforms uncalibrated (0.330); calibration homogeneity boundary condition |

| | | | |
|---|---|---|---|
| | `calibration` | calibration vs uncalibrated | confirmed; geo calibration deferred |
| `824da3d` | — | Freeze calibration architecture in ADR-005 (docs-only commit) | Monetary-only calibration policy formalised; geopolitical calibration explicitly deferred pending horizon-stratified empirical gap estimation |
| `993bf11` | `v0.5.8-monte-carlo` | Block-bootstrap robustness test on current best stack (B=500, block=20 days) | Observed Sortino 0.330; bootstrap mean 0.454; 95% CI [-0.651, 1.806]; p=0.542; directionally credible, not statistically settled; bootstrap mean exceeds observed, confirming architecture is not ceiling-constrained at point estimate |
| `1981ef6` | — | Freeze Paper 2 framing (docs-only commit) | Infrastructure claim, expansion-as-informational-sensitivity reframe, and regime-class generalisation boundary condition locked into paper draft |

---

## Appendix B: Current Architecture Specification

The following is the full machine-readable specification of the current best architecture as of tag `v0.5.8-monte-carlo`. This appendix is intended to be a complete drop-in reference for replication.

text

```
# Cassandra-Risk V5 Architecture Specification
# Tag: v0.5.8-monte-carlo | Commit: 993bf11
```

```
UNIVERSE:
  total_events: 34
  sources:
    - polymarket_historical: 28 events
    - metaculus_manual: 6 events

STRUCTURAL_THEME_POLICY:
  monetary_policy:
    becker_calibration: true
    efficiency_gap: 0.0017
    longshot_compression: true  # P < 0.20 or P > 0.80
    top_5_removal: true          # remove 5 highest cumulative-
hazard contracts
    bucket_cap: 0.30             # max 30% of total hazard from
this theme
  geopolitical:
    admission_floor_volume_usd: 500_000
    binary_resolution_required: true
    macro_relevant_required: true
    bucket_cap: 0.25
    becker_calibration: false   # deferred per ADR-005
  electoral:
    standard_admission: true
    calibration: none
  fiscal_debt:
    standard_admission: true
    calibration: none
  systemic_credit:
    standard_admission: true
```

```
    calibration: none
  trade_technology:
    standard_admission: true
    calibration: none

EFFICIENCY_GAPS (reference only — applied to monetary only):
  monetary_policy:   0.0017
  electoral:         0.0102
  fiscal_debt:       0.0102
  systemic_credit:   0.0102
  trade_technology:  0.0269
  geopolitical:         0.0732   # not applied — theoretically
estimated, deferred

RSI_FORMULA:
  RSI_t = 1 / (1 + H_t)
  H_t = sum_i(lambda_i * d_i_t * P_i_t)
  d_i_t = exp(-lambda_i * tau_i_t / 30)
  Position_t = RSI_t * 100

BACKTEST:
  asset: SPY
  period: 2020-01-01 to 2025-01-10
  frequency: daily
  slippage_bps: 5
  cash_return: FRED TB3MS
  leverage: none

CURRENT_BEST_PERFORMANCE:
```

```
sortino:        0.330
cagr:           7.24%
daily_mdd:     -33.72%
downside_dev:   0.1431
cvar_95:       -0.0313
worst_mo_mdd:  -0.2832
avg_position:  72.54%
```

---

## Appendix C: Reviewer-Safe Claims

The following claims are supported by the evidence in this paper without overreach. Each is bounded precisely.

**C1.** Expansion from 9 to 38 events degrades Sortino from 1.159 to 0.231. The degradation is causally attributable to monetary-policy concentration, not to broad event-universe contamination.

**C2.** Removing the top-5 cumulative-hazard monetary contracts improves Sortino from 0.231 to 0.294. Crude family compression worsens Sortino to 0.211. The directional asymmetry is explained by the mechanism through which family compression concentrates bias rather than dispersing it.

**C3.** Becker efficiency-gap calibration applied to the monetary-policy bucket lifts Sortino from 0.231 to 0.241 and is near-additive with structural fixes. The lift is entirely a CAGR-numerator effect. Downside deviation and $CVaR_{95}$ are unchanged.

**C4.** The three-layer monetary stack (Becker + top-5 removal + 30% cap) improves Sortino by 39.8% and CAGR by 1.38pp relative to the naive V5 expansion baseline, with $CVaR_{95}$ and worst monthly drawdown effectively unchanged across all five stack configurations.

**C5.** Governed geopolitical admission adds +0.007 Sortino over the monetary-only stack. Flat and sub-bucket geopolitical calibration both degrade performance relative to uncalibrated admission. The asymmetry is predicted by the domain-by-horizon interaction literature.

**C6.** Block-bootstrap Monte Carlo (B = 500, block = 20 days) on the current best stack yields observed Sortino 0.330, bootstrap mean 0.454, 95% CI [-0.651, 1.806], p = 0.542. The strategy is directionally credible, not statistically settled.

**C7.** The bootstrap mean Sortino (0.454) exceeds the observed result (0.330), confirming the architecture is not ceiling-constrained at the observed point estimate and that the realised path was a slightly below-median draw from the strategy's own bootstrap distribution.

---

# Appendix D: Open Research Agenda

This appendix records the explicit gaps that Paper 2 leaves open, in priority order, so that readers can assess what the next stage of the research program targets.

**D1 — Comparative bootstrap (near-term).** Run the same block sequence simultaneously across Buy & Hold, Volatility Targeting, V5 baseline, and current best stack. Compute win-rate matrices and CI on Sortino differential. This is the stronger distributional robustness test that single-strategy CIs cannot provide.

**D2 — Kelly-optimal position sizing (v0.5.9).** Apply Kelly-optimal weighting using the Becker-corrected $PM/PY$PM/PY ratio per contract to target the downside deviation denominator. The current architecture has no lever on this dimension. Kelly and Becker are complementary fixes on orthogonal Sortino dimensions.

**D3 — Kalshi ingestion and cross-platform validation (v0.6.0, Paper 3).** Build the Kalshi REST pipeline (RSA-PSS auth) and test the near-zero efficiency gap hypothesis on Fed rate decision markets. This provides (a) a structural robustness check against a genuinely independent signal source and (b) the cleanest possible empirical calibration environment for the monetary bucket — which already has the highest empirical anchor quality of any theme.

**D4 — Geopolitical horizon calibration (deferred).** Horizon-stratified empirical gap estimation for geopolitical sub-buckets requires contract-level resolution history stratified by sub-bucket × time-to-resolution. This is not feasible from current Polymarket snapshot data. It becomes feasible once six to twelve months of live resolved contracts are collected and tagged.

**D5 — Regime-class generalisation test.** No current test addresses whether the architecture retains signal properties under a genuinely new regime type. The Kalshi ensemble (D3) is a partial answer via source independence. A fuller answer requires either extending the backtest sample or constructing a synthetic out-of-sample stress scenario using the multi-asset hazard sensitivity matrix from Paper 1.

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